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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 8

Title: Marketing Campaign Performance Dashboard

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Marketing Campaign Performance Dashboard

Problem Statement

You are a Marketing Analyst in a digital marketing agency. Multiple marketing campaigns have been executed across different channels. Management wants to evaluate campaign performance based on impressions, clicks, conversions, campaign cost, revenue, and Return on Investment (ROI). The objective is to develop an interactive Power BI dashboard that provides meaningful insights into campaign effectiveness, channel performance, customer engagement, and marketing profitability to support data-driven marketing decisions.

Dataset Used

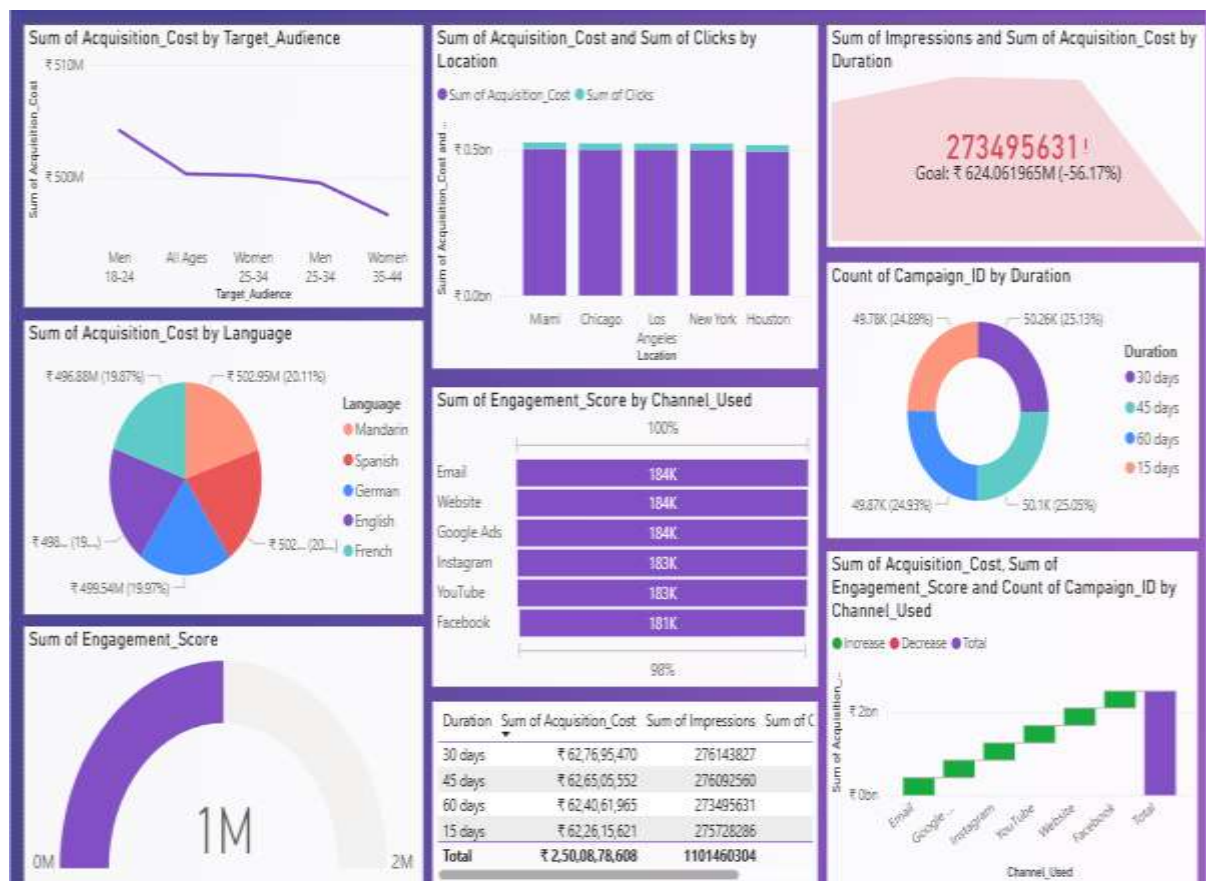
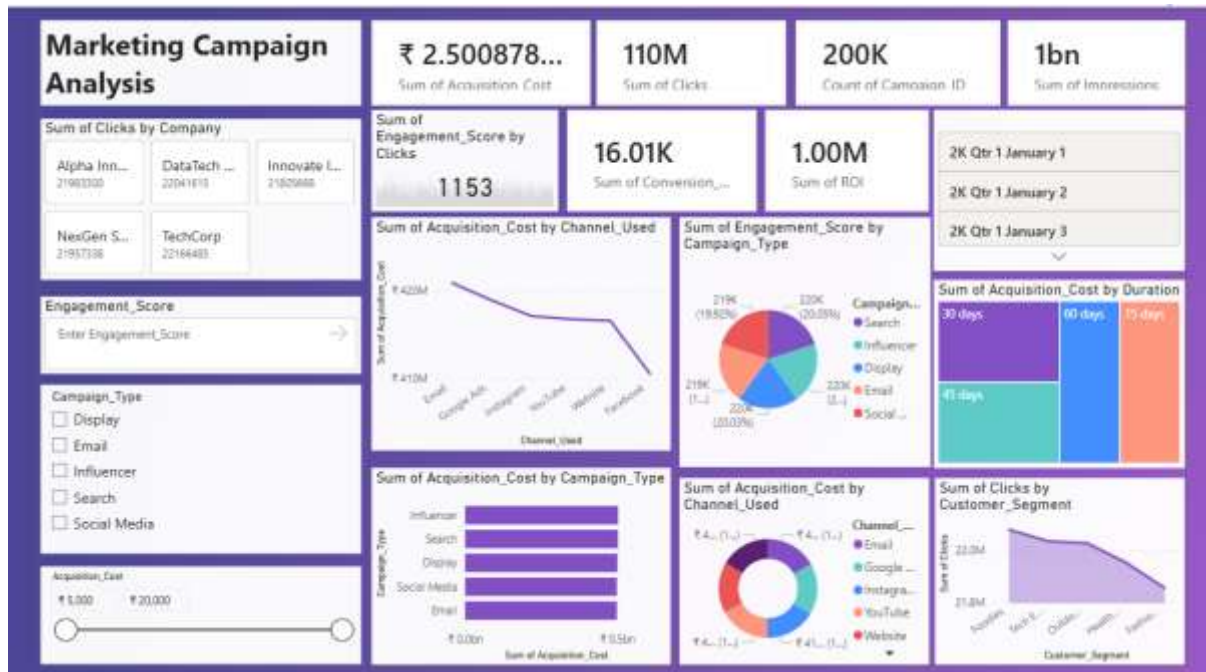
Marketing Campaign Performance Dataset (Kaggle)

Methodology / Steps Followed

1. Downloaded the **Marketing Campaign Performance Dataset** from Kaggle.
2. Imported the dataset into **Power BI Desktop** using the **Get Data** option.
3. Examined the dataset structure and verified column names, data types, duplicate records, and missing values using **Power Query Editor**.
4. Cleaned and transformed the data wherever required before loading it into Power BI.
5. Created calculated measures using **DAX** to analyze important marketing KPIs such as Total Impressions, Total Clicks, Total Conversions, Total Campaign Cost, Total Revenue, Click-Through Rate (CTR), Conversion Rate, and ROI.
6. Created **KPI Cards** to display the overall performance of marketing campaigns.
7. Developed interactive visualizations including **Bar Charts, Column Charts, Line Charts, Donut Charts, Treemap, Scatter Charts, and Tables**.
8. Analyzed campaign performance across different marketing channels, campaign types, customer segments, and time periods.
9. Added **Slicers** for Campaign Type, Channel, Date, Region, and Customer Segment to enable dynamic filtering.
10. Compared marketing expenditure with revenue generated to identify profitable and underperforming campaigns.
11. Analyzed impressions, clicks, conversions, and ROI to understand customer engagement and campaign effectiveness.

12. Applied appropriate formatting, titles, colors, and layout techniques to create a professional and user-friendly **Marketing Campaign Performance Dashboard**.

Output



Results

The Marketing Campaign Performance dataset was successfully imported, cleaned, and analyzed using Power BI.

KPI cards provided a quick overview of important campaign metrics such as impressions, clicks, conversions, revenue, and ROI.

Channel-wise analysis helped identify which marketing platforms generated the highest customer engagement.

CTR analysis measured how effectively campaigns converted impressions into customer clicks.

Conversion rate analysis helped evaluate the ability of campaigns to convert interested users into customers.

ROI analysis identified profitable campaigns and campaigns requiring improvement.

Revenue and campaign cost comparison helped management understand marketing profitability.

Time-based analysis revealed trends and changes in campaign performance.

Interactive slicers enabled users to filter campaign data dynamically based on channel, campaign type, region, and time period.

Business Insights

Marketing campaigns generate different levels of impressions and engagement depending on the channel used.

Campaigns with high impressions do not always generate high conversions, highlighting the importance of analyzing the complete marketing funnel.

Click-Through Rate (CTR) helps identify campaigns that successfully attract customer attention.

Conversion Rate provides insights into how effectively customer clicks are transformed into successful conversions.

Certain marketing channels generate higher revenue and ROI, making them more suitable for future marketing investments.

Campaigns with high costs but low revenue can be identified and optimized to reduce unnecessary marketing expenditure.

Comparing campaign cost and revenue helps management measure the profitability of individual marketing campaigns.

Time-based analysis helps identify periods with stronger campaign performance and customer engagement.

Campaign type analysis enables marketers to understand which promotional strategies produce better results.

Interactive filters allow management to perform detailed campaign analysis and make data-driven marketing decisions.

ROI is an important performance indicator because it helps determine whether marketing investments are generating sufficient returns.

The dashboard helps marketing teams allocate budgets efficiently toward high-performing channels and campaigns.

Conclusion

This experiment demonstrated the development of an interactive **Marketing Campaign Performance Dashboard using Microsoft Power BI**. The dashboard integrates multiple KPI cards, DAX measures, charts, tables, and interactive slicers to analyze important marketing metrics such as impressions, clicks, conversions, campaign cost, revenue, Click-Through Rate, Conversion Rate, and Return on Investment.

The dashboard provides a comprehensive view of campaign performance across different marketing channels and campaign types. It helps identify high-performing campaigns, evaluate customer engagement, compare marketing costs with generated revenue, and measure overall campaign profitability. The insights generated from the dashboard can support marketing managers in optimizing campaign strategies, improving budget allocation, and making effective data-driven business decisions.

Learning Outcomes

- ✓ Understand the importance of Marketing Analytics in business decision-making.
- ✓ Import and prepare a Marketing Campaign Performance Dataset in Power BI.
- ✓ Perform data cleaning and validation using Power Query Editor.
- ✓ Create DAX Measures for important marketing KPIs.
- ✓ Calculate Total Impressions, Clicks, Conversions, Revenue, and Campaign Cost.

- ✓ Calculate Click-Through Rate (CTR) and Conversion Rate.
- ✓ Calculate Return on Investment (ROI) to evaluate campaign profitability.
- ✓ Create KPI Cards to summarize overall marketing performance.
- ✓ Build Bar Charts, Column Charts, Donut Charts, Line Charts, Treemaps, and Scatter Charts.
- ✓ Analyze campaign performance across different marketing channels.
- ✓ Compare campaign costs with generated revenue.
- ✓ Identify high-performing and underperforming marketing campaigns.
- ✓ Analyze customer engagement using impressions, clicks, and conversions.
- ✓ Implement interactive slicers for dynamic dashboard filtering.
- ✓ Apply formatting and visualization techniques to create a professional Power BI dashboard.
- ✓ Gain practical knowledge of using Business Analytics and Power BI for marketing decision-making.

GitHub Link

https://github.com/lavanyae4319/Data_Analytics_Assignments